



ABSTRACT

1. Title: HARVIANA: A GIS-INTEGRATED DECISION SUPPORT SYSTEM FOR VISUALIZING VEGETABLE PRODUCTION TRENDS IN BENGUET
 - 1.1. Total No. of Pages: 152
 - 1.2. Text No. of Pages: 107
2. Authors: BATERINA, LORD AFLREY T.
DELA CRUZ, JOHNDOE AMIEL I.
TAGALOG, JOJEMAY M.
3. Type of Document: Capstone Project
4. Type of Publication: Unpublished
5. Accrediting Institution: University of the
Cordilleras, Gov. Pack Rd.
Baguio City
6. Keywords: Harviana, Geographic Information System,
Agricultural Decision-Support, Vegetable
Production, Benguet
7. Abstracts:

7.1 Rationale/Background of the Study

Vegetable production is essential to Benguet's economy, agricultural livelihoods, and contribution to the regional food supply. However, crop plans, damage reports, and actual-harvest records can be difficult to consolidate when information is handled through separate or uncoordinated processes. Fragmented and delayed records may affect report validation, production monitoring, and the availability of timely information for municipal agricultural planning. These limitations also make it difficult to compare expected production with actual harvest outcomes and identify possible changes in vegetable supply. Farmers therefore need an organized way to manage crop activities and submit farm records, while agricultural offices require updated, validated, and location-based information for monitoring and decision-making. To address these needs, the researchers developed Harviana, a GIS-integrated decision-support system connecting Farmers, LGU



Validators, and DA Administrators through submission, validation, monitoring, forecasting, and reporting.

7.2 Summary

This study aimed to design and develop Harviana: A GIS-Integrated Decision Support System for Visualizing Vegetable Production Trends in Benguet. Specifically, the researchers sought to:

1. to identify the information requirements of proposed system;
2. to determine the architectural framework of the proposed system;
3. to identify the features of the proposed system; and
4. to measure the extent of usability of Harviana among participating Farmers and DA Administrators.

The researchers used Scrum through five Sprints: Requirements Gathering and Planning, GIS Visualization Module Development, Core Feature Development, Machine-Learning Production Estimation and Supply Forecast Integration, and Testing and Evaluation. This process supported iterative development and refinement.

7.3 Findings

Based on the data gathered and analyzed in relation to the objectives of the study, the researchers identified the following findings:

1. Harviana's information requirements consist of Farmer crop-planning records, damage reports, actual-harvest records, LGU validation data, DA monitoring and reporting information, and the technical requirements necessary to support forecasting, GIS-based visualization, validation, and decision-support functions.

2. The "4+1" View Model described Harviana's scenario, logical, process, development, and physical views, including its users, structure, workflows, software components, and Railway deployment with its machine-learning service and Railway PostgreSQL.

3. Harviana provides role-based features for its three primary user groups. Farmers can create crop plans, manage farming calendars, generate machine-learning-based production estimates, receive crop recommendations, submit damage reports, record actual harvests, and access GIS maps and weather information. LGU Validators can



review, validate, approve, and provide correction notes for Farmer-submitted reports. DA Administrators can manage agricultural records, monitor dashboards, visualize GIS-based production information, access municipal supply forecasts, review prediction analytics, generate reports, and export data. These features establish an integrated workflow that supports agricultural monitoring and decision-making in Bengu

4. The 132 valid Farmer respondents obtained a mean SUS score of 71.55, interpreted as "OK to Good" and "Marginal to Acceptable," indicating that Harviana possesses an acceptable level of usability from the perspective of farmer users. Meanwhile, the 18 DA Administrator respondents obtained an overall USE mean score of 6.32, interpreted as "Strongly Agree," indicating a highly positive evaluation in terms of usefulness, ease of use, ease of learning, and satisfaction. These results demonstrate that Harviana is usable and acceptable for its intended users.

7.4. Conclusions

Based on the findings of the study, the researchers arrived at the following conclusions:

1. The identified information requirements provide the records needed for crop monitoring, validation, forecasting, and reporting.

2. The "4+1" View Model provides an organized framework for describing Harviana's users, structure, processes, components, and deployment.

3. Harviana successfully implemented role-based features for Farmers, LGU Validators, and DA Administrators. Farmers were provided with crop planning, calendar management, machine-learning-based production estimation, damage reporting, harvest recording, crop recommendations, and access to GIS maps and weather information. LGU Validators were provided with report review, validation, and correction functions, while DA Administrators were provided with dashboard monitoring, GIS visualization, municipal supply forecasting, prediction analytics, report generation, and export capabilities. These features establish an integrated workflow that supports agricultural monitoring and decision-making across stakeholder groups.

4. Harviana demonstrated an acceptable level of usability among participating Farmers and a highly



positive usability evaluation among DA Administrators. The farmer respondents obtained a SUS score interpreted as "OK to Good" and "Marginal to Acceptable," while the DA Administrators strongly agreed that the system was useful, easy to use, easy to learn, and satisfactory. These results indicate that Harviana is suitable for its intended users, although improvements to the farmer interface may further enhance usability for first-time users.

7.5 Recommendations

Based on the conclusions of the study, the researchers recommend the following:

1. Regularly review Harviana's information and validation requirements with its user groups to maintain complete and relevant agricultural records.
 2. Maintain the database, machine-learning service, security controls, backups, and Railway deployment for reliable and secure operation.
 3. Expand crop coverage and validated datasets, improve user guidance, and support Farmers in areas with limited connectivity.
 4. Include LGU Validators and participants from other Benguet municipalities in future usability evaluations to improve the interface and validation workflow.
-